

VPIN: calculation, usage and backtests

A practical guide to this Binance minute-data implementation

Study snapshot: 2020-01-01 to 2026-09-20 00:00:00 UTC (end exclusive).

Status: data unavailable.

1 What the indicator measures

Volume-Synchronized Probability of Informed Trading (VPIN) summarizes absolute buy/sell imbalance in volume-based observations. A volume clock advances when enough trading volume has accumulated, rather than at equally spaced times. Under the assumptions behind PIN models, persistent imbalance can be associated with informed activity and adverse selection faced by liquidity providers [1, 3]. Imbalance alone does not identify a trader's information.

This project constructs an *approximation* from completed one-minute Binance spot candles. It uses the exchange's aggregate taker-buy volume, rather than an inferred tick rule. Minute totals do not reveal trade ordering. High VPIN can accompany buying as well as selling; it supplies no direction.

1.1 Data and adaptive bucket targets

All dates and candle boundaries are UTC. For minute m , let v_m be total base-asset volume and b_m taker-buy base volume. Sell-initiated volume is $s_m = v_m - b_m$. For BNBUSDT these volumes are in BNB; prices are USDT per BNB. The API fields are documented by Binance [5]. The implementation requires finite positive prices and $0 \leq b_m \leq v_m$.

For UTC day d , the target for a *new* bucket is

$$A_d = \frac{1}{L} \sum_{\ell=1}^L \sum_{\substack{m \text{ on complete} \\ \text{UTC day } d-\ell}} v_m, \quad V_d = \frac{A_d}{k}, \quad (1)$$

where $L = \text{adv_lookback_days}$ and k is the target bucket count per day. Only preceding complete days enter the average. Initial calibration rows have no targets. A positive `bucket_size_base` override supplies a fixed target and bypasses calibration.

Once a bucket starts, its target stays fixed until it fills, including across midnight. Later buckets use the new day's target. Thus k is an activity target based on recent volume, not a guarantee of exactly k observations every day.

1.2 Filling and normalizing

When f units from a minute fill part of a bucket, assigned buys and sells are $f b_m/v_m$ and $f s_m/v_m$. Repeat until the minute is consumed; retain the unfinished bucket for the next minute and omit the final incomplete tail. For the most recent n full buckets,

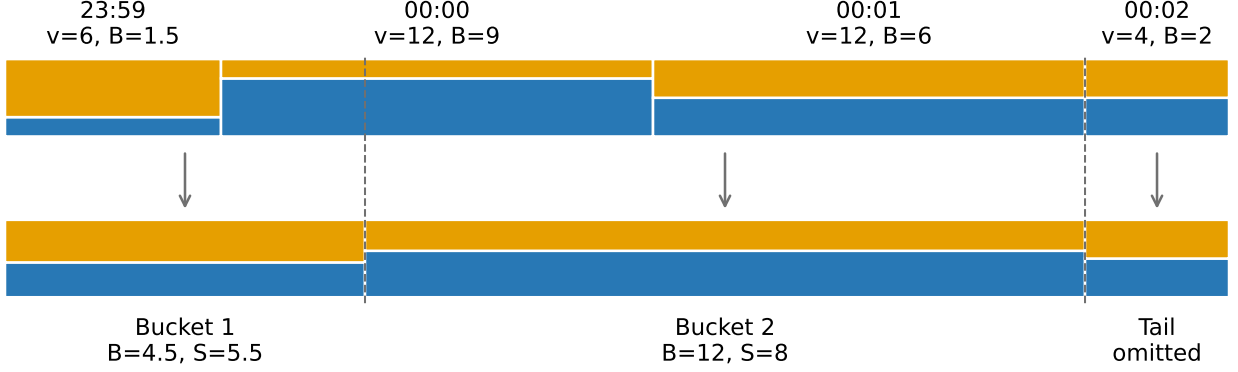
$$\text{VPIN}_i = \frac{\sum_{j=i-n+1}^i |B_j - S_j|}{\sum_{j=i-n+1}^i (B_j + S_j)}. \quad (2)$$

The denominator is actual rolling volume. If all bucket sizes equal V , it reduces to nV , the fixed-volume formula in [1, section 19.5.2, p. 292]. Changing daily targets is this project's extension of that setting.

2 From minutes to volume buckets

Consider these base volumes across midnight. The day change alters targets only for buckets that have not yet started.

Blue: buys Orange: sells Width: base volume



Worked example. Each upper rectangle is a minute's buy/sell mixture; vertical colors do not imply trade ordering. Dashed boundaries show where the volume clock splits a minute. The new-bucket target changes from 10 to 20 at midnight. Bucket 1 takes six units from 23:59 and four from 00:00: $B_1 = 1.5 + 4(9/12) = 4.5$ and $S_1 = 5.5$. It keeps its original target of 10. Bucket 2 takes the remaining eight units from 00:00 and all twelve from 00:01: $B_2 = 8(9/12) + 6 = 12$ and $S_2 = 8$. With $n = 2$,

$$\text{VPIN} = \frac{|4.5 - 5.5| + |12 - 8|}{10 + 20} = \frac{5}{30} \simeq 0.1667.$$

The four final units remain unfinished and do not enter this estimate.

2.1 When a value becomes usable

A bucket's **time** is the closing timestamp of its final contributing minute plus one millisecond. The example buckets become available at 00:01 and 00:02, respectively. Recording their starting times would introduce future information into earlier signals.

Several buckets may complete within the same minute. Construction order is preserved; the simulator uses the last completed bucket for that minute. Bucket **price** is the final candle's close for plotting, not a median trade price or a trade execution price.

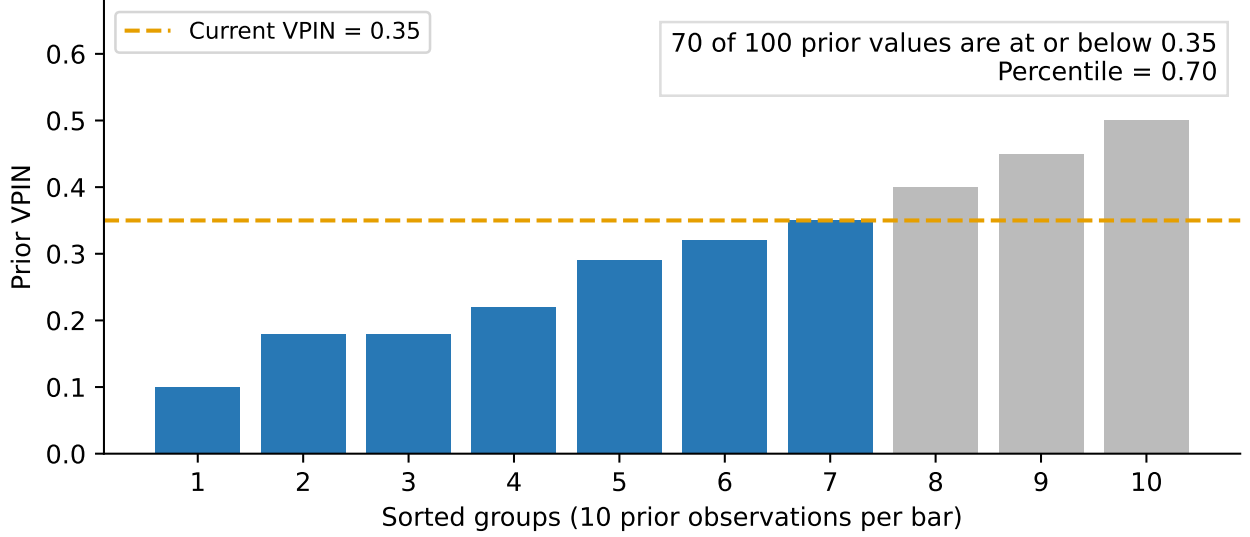
2.2 The minute-data limitation

If a minute contains 100 buys followed by 100 sells, trade-sequenced buckets of size 10 would be one-sided. Proportional splitting of the balanced minute instead produces balanced buckets: VPIN 1 versus 0. Exchange aggressor totals classify the *minute*, but cannot recover its missing sequence.

3 Percentiles and performance measures

Let $\mathcal{H}_i$ contain preceding valid VPIN observations in the configured D -day window. The current observation is excluded; ties count:

$$F_i = \frac{1}{|\mathcal{H}_i|} \sum_{j \in \mathcal{H}_i} \mathbf{1}\{\text{VPIN}_j \leq \text{VPIN}_i\}. \quad (3)$$



Reading the rank. Each bar represents ten prior observations at the shown value. Seven blue bars give 70 observations at or below the current VPIN of 0.35, so $F_i = 70/100 = 0.70$. Ties are included. The current value is compared with the reference; it is not added to its own denominator.

The window excludes its left boundary. Earlier bucket rows with the same availability timestamp retain their order. CDF is unavailable until D days have elapsed since the first valid VPIN and at least 100 prior values are present. A constant reference also yields no CDF.

CDF 0.99 places the observation near the upper tail of its reference history. It is not a 99% crash probability or a calibrated probability that a trader is informed.

The drawing is a constructed teaching example, not market data. It assumes the required time history is present. It shows how the rank is calculated, not whether a rank predicts a useful trading decision.

3.1 Returns, drawdown, and Calmar

Equity E_t is cash plus asset quantity times the minute close. Two successive doublings take wealth from 1 to 4: a 300% return. A decline from 2 to 1 is a 50% drawdown. Including initial capital in the peak,

$$DD_t = 100 \left(\frac{E_t}{\max_{u \leq t} E_u} - 1 \right).$$

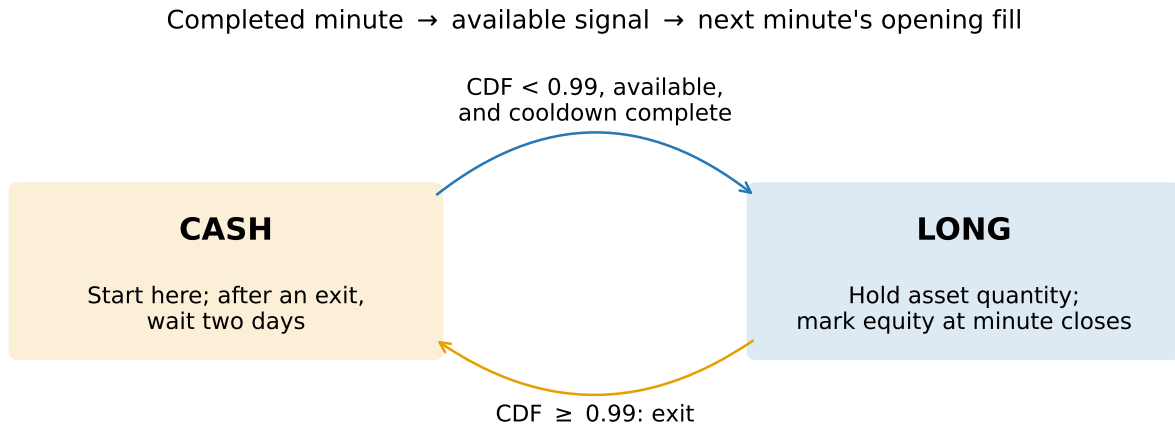
Over d calendar days, compound annual growth rate and the selection score are

$$\text{CAGR} = 100 \left[\left(\frac{E_T}{E_0} \right)^{365.25/d} - 1 \right], \quad \text{Calmar} = \frac{\text{CAGR}}{|\min_t DD_t|}.$$

Costs enter equity before these metrics are calculated. Zero drawdown or nonfinite annualization makes a candidate ineligible.

4 Basic strategy and usage

This is a long-or-cash risk-exit experiment, not a market-making implementation.



Missing CDF: keep the current position. End of test: liquidate holdings.

Start each scoring interval with unit cash. The two-day cooldown starts at the actual exit, not when a bucket begins. At the end, remaining holdings are sold at the final candle close; this liquidation is not a signal exit. A signal formed when a candle finishes can first act at the next candle's open. Exit takes precedence over re-entry, preventing a same-time round trip.

4.1 Configuration

Setting	Study configuration
Symbol / requested history start	BNBUSDT / 2020-01-01
Baseline buckets/day / VPIN window	252 / 10
ADV / CDF lookback	90 / 90 days
Fixed bucket override	None: adaptive targets
Fee / slippage per side	10 / 2 basis points
Exit threshold / cooldown	0.99 / 2 days

The study does not silently change application defaults. The baseline 252/10 settings are provisional. Costs are adjustable assumptions, not the user's actual exchange fee tier. Cash earns no interest.

4.2 Commands

Use Python 3.10+ from the project root:

```
pip install pandas numpy aiohttp matplotlib pyarrow tqdm optuna
python -m unittest -q test_vpin
python market_data.py
python vpin_calculator.py
python vpin_backtest.py
python vpin_optuna_optimize.py
```

Edit `config.json` for the instrument and a valid data interval. The downloader repairs missing ranges and refreshes overlapping closed candles. Unresolved gaps stop analysis. Old timezone-naive caches are rebuilt because historically truncated candles cannot be identified reliably.

5 Search design and empirical results

5.1 Training and held-out evaluation

Both methods use the same data, warmup, simulator, costs, and scoring dates. The common warmup uses 20 buckets/day and a 350-bucket window, or more demanding configured values. Candidates without usable history at the shared start are rejected.

After warmup, the first 80% of elapsed time is training and the last 20% is held out. The protocol requests 143 grid combinations and 100 Optuna TPE trials (seed 42, 20 startup trials), searching 20–350 buckets/day and windows 10–350. Eligibility requires at least 30 training signal exits and finite Calmar. Only training Calmar selects a winner.

The held-out comparison includes each method’s winner and the current 252/10 setting, deduplicating identical configurations. Each starts with fresh cash while retaining preceding indicator history. Buy-and-hold uses identical dates and entry/terminal cost assumptions. A poor test result does not cause another configuration to be selected.

5.2 Results unavailable under the requested constraints

Study status: data unavailable. The requested history contains 2,347 missing one-minute candles across 15 intervals. All 15 direct API rechecks returned zero candles. Checksum-verified official archives for February 2020 and March 2023 contain the same omissions in those months. The searches were not started; no historical winner is claimed.

No winning configuration or performance table is asserted. Old avoided- drawdown statistics and synthetic tests are not empirical substitutes. Coverage must pass before either full search supplies results. Any change to the study interval must be explicit and recorded.

5.3 Interpreting results

Selling during strong buying can miss upside. Judge cost-adjusted test return and drawdown together with the benchmark. A brief dip while in cash does not prove that an exit was beneficial; the former cumulative avoided-drawdown score was removed for this reason.

Minute-close equity omits intramminute extremes. Fixed-slippage opening fills do not model spread dynamics, order-book depth, impact, latency, or trading halts. Repeatedly tuning after inspecting the test period would consume its independence. No uncertainty interval or prediction accuracy is established by Calmar alone.

6 Book context, verification, and reproduction

These are original summaries of relevant ideas, not reproductions of the source books.

PIN and volume clocks. López de Prado [1, section 19.5.2, p. 292] relates normalized imbalance to PIN under arrival-model assumptions. The fixed-volume algebra supports the special case of Eq. (2); the discussion also records conflicting findings about volatility prediction. Correct arithmetic alone does not establish forecasting value.

Calibration is a separate result. The futures study summarized in [1, section 22.6.5, pp. 346–347] covers approximately 100 contracts over 67 months. Its example settings include 200 buckets/day, 30 *bars per bucket*, one-day VPIN support, 0.1-day events, median trade prices, Student- t bulk classification ($\nu = 0.1$), and a 0.99 CDF threshold. The underlying report separates bar construction from the number of buckets used in VPIN [4]. Thirty bars per bucket is not a 30-bucket rolling window; this calibration does not validate the project’s adaptive minute-based crypto settings. That study evaluates volatility events rather than this long/cash strategy’s returns.

Other context. Jansen [2, pp. 38–39] explains volume bars, illustrates timestamps at the last constituent trade, and discusses dollar bars when price changes affect comparisons. Cartea, Jaimungal, and Penalva [3, p. 70] discuss PIN alongside information asymmetry and market quality. Neither supplied a separate VPIN calibration used here. The scanned Cartea volume was inspected through its index and related pages, not exhaustive OCR.

Section 19.7 of [1, pp. 295–296] concerns predictive losses and microstructural information, not this empirical CDF. A high/low midpoint is not a median trade price. With supplied aggressor totals, the project’s plotted bucket prices do not enter Eq. (2).

6.1 Reproduction

The manifest records configuration, source/data hashes, versions, and coverage outcome. A completed study also records scoring dates and seed, with full training trials and selected held-out metrics in CSV files. Rebuilding generates tables from that evidence and verifies its hashes without rerunning optimization. Text fingerprints normalize CRLF line endings to LF; binary fingerprints are byte-exact. Rebuild or refresh with:

```
python docs/build_vpin_guide.py
python docs/build_vpin_guide.py --run-backtests
```

Use `-start-date YYYY-MM-DD` and `-end-date YYYY-MM-DD` with the second command to set an explicit study interval (end exclusive) without changing `config.json`. Tests cover independent arithmetic, data boundaries, causality, execution, costs, and selection. Grid/Optuna agreement checks their integration rather than independently proving the numerical model.

References

- [1] M. López de Prado. *Advances in Financial Machine Learning*. Wiley, 2018. Sections 19.5.2, 19.7, and 22.6.5.
- [2] S. Jansen. *Machine Learning for Algorithmic Trading*, 2nd ed. Packt, 2020, pp. 38–39.
- [3] Á. Cartea, S. Jaimungal, and J. Penalva. *Algorithmic and High-Frequency Trading*. Cambridge University Press, 2015, p. 70.
- [4] K. Wu, E. W. Bethel, M. Gu, D. Leinweber, and O. Rübel. *A Big Data Approach to Analyzing Market Volatility*. LBNL-6382E, 2013. [Public technical report](#).
- [5] Binance. *Spot REST API: Market data endpoints*. [Official kline documentation](#). Accessed 20 September 2026.